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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (CSE/ISE) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS503 / 20CS502 / 21IS503 - DATABASE MANAGEMENT SYSTEMS
17CS502 / 17IS502 - RELATIONAL DATABASE MANAGEMENT SYSTEMS
20IS502 - DATABASE SYSTEM MODELS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the characteristics of database approach.	10	L*2	1	1
	b) Write an ER diagram to represent the UNIVERSITY database schema and indicate all key and cardinality constraints.	10	L3	1	2
2.	a) Define the following terms: i) Data Model ii) Canned transaction iii) Schema iv) Data Manipulation Language (DML) v) Instance	10	L2	1	1
	b) Briefly discuss the different type of update operations on relational database. Show an example of a violation of the referential integrity in each of the update operations.	10	L3	2	1
3.	a) Summarize the steps involved in converting the ER constraints to relational schema.	10	L2	1	2
	b) Consider the following schema. SAILOR (<u>SID</u> , SNAME, RATING, AGE) BOAT (<u>BID</u> , BNAME, COLOR) RESERVE (<u>SID</u> , BID, DAY) Specify the following queries in relational algebra. i) Retrieve the sailor names that have reserved red and green boats ii) Retrieve the colors of boats reserved by Raj iii) Retrieve the SIDs of sailors with age over 20, who have not reserved a red boat. iv) Retrieve the names of sailor who have reserved all boats.	10	L3	2	2
Unit – II					
4.	a) With real world example, explain the following. i) Correlated nested queries ii) Stored procedures iii) Schema change statements in SQL	10	L2	3	2
	b) Discuss informal guidelines to determine the quality of relations schema design with a suitable example.	10	L2	3	3
5.	a) What is the need for normalization? Explain 1NF, 2NF and 3NF with examples.	10	L2	4	1
	b) What is functional dependency? Write an algorithm to find minimal cover for set of functional dependencies. Construct minimal cover m for a set of FD E: {B→A, D→A, AB→D}	10	L3	4	3

P.T.O.

6. a) For the schema given in Q. No. 3(b), specify the following queries in SQL.
- Find the names of sailors who have reserved a red or a green boat
 - Find the names of sailors who are older than the oldest sailors with a rating of 10
 - Find sailors whose rating is better than some sailor called 'Ramesh'
- b) What are views? Explain the specification and implementation of views in SQL.

10	L3	3	2
10	L2	3	1

Unit – III

7. a) Why concurrency control is needed? Demonstrate with an example. 12 L3 5 1
- b) Describe the ACID properties of database transaction. 08 L2 5 1
8. a) Discuss Two-phase locking technique for concurrency control. 10 L2 5 2
- b) Discuss CRUID operations in MONGODB 10 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
